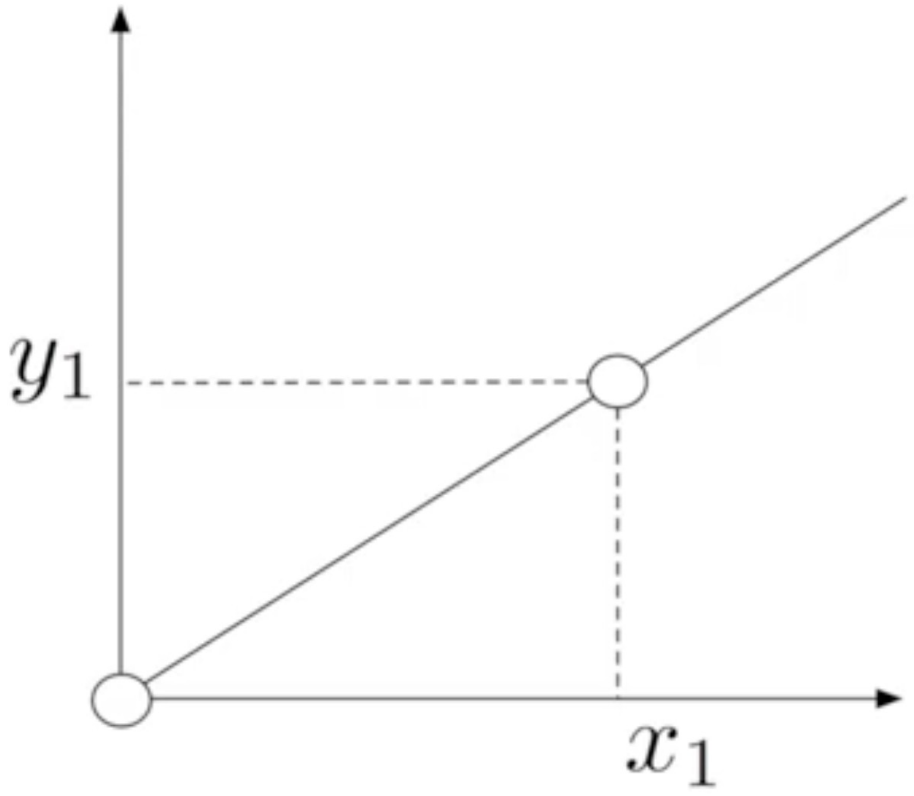
UNIVERSITAT POLITÈCNICA DE CATALUNYA BARCELONATECH

GIA - IAA Introducció a l'Aprenentatge Automàtic

4. Lab: Linear Regression

Jordi Luque
jordi.luque.serrano@upc.edu

Linear Regression

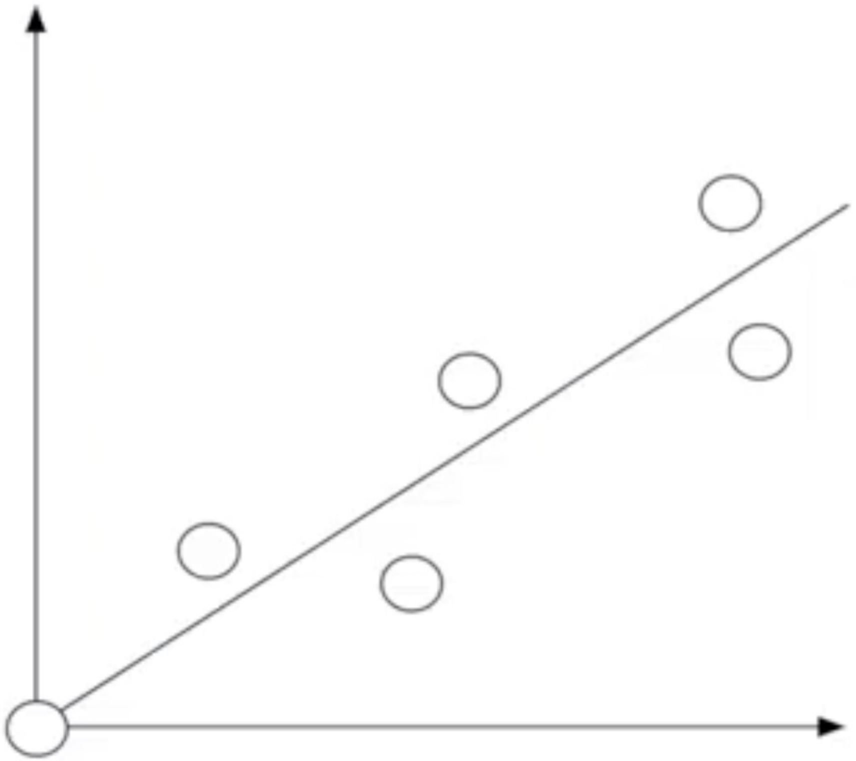


$$y = ax$$

$$a = \frac{y_1}{x_1}$$

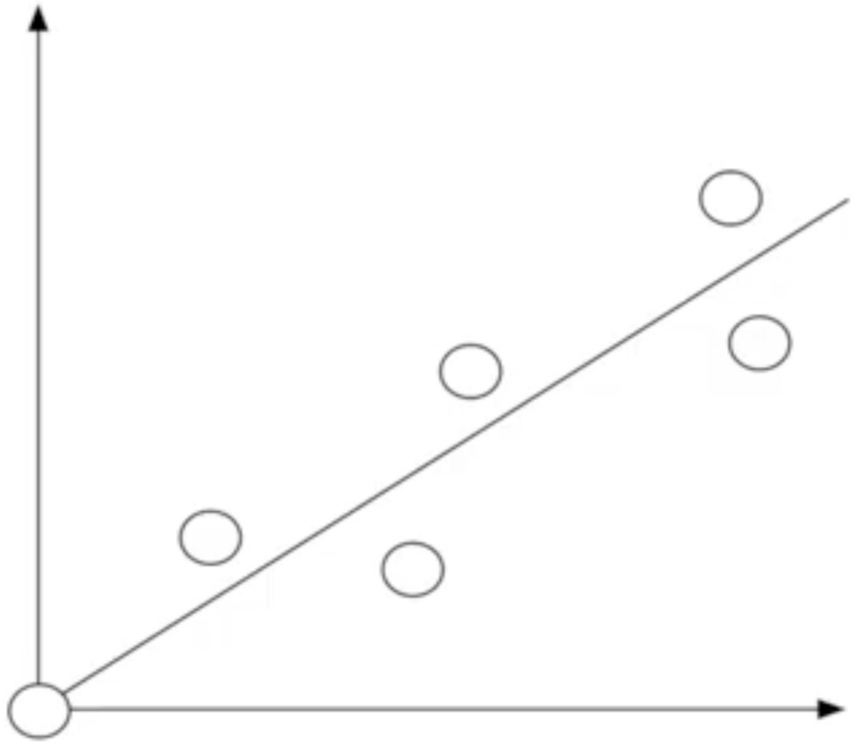
Linear Regression

$$y = ax$$



$$\mathbf{x} = x_1, x_2, \ldots, x_n$$
$$\mathbf{y} = y_1, y_2, \ldots, y_n$$
$$a = ??$$

Linear Regression



$$y = ax$$

$$\mathbf{x} = x_1, x_2, \ldots, x_n$$

$$\mathbf{y} = y_1, y_2, \ldots, y_n$$

$$a = \frac{\mathbf{y}\mathbf{x}'}{\mathbf{x}\mathbf{x}'}$$

Correlation Analysis

- Correlation analysis illustrates the degree of a relationship between two variables, i.e. how closely they are correlated. The result is always between -1.0 and 1.0, where 1.0 denotes the highest positive correlation, -1.0 stands for the highest negative correlation (explained below), and 0 means no correlation

$$r_{xy} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

where

- n is sample size
- x_i, y_i are the individual sample points indexed with *i*
- $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$ (the sample mean); and analogously for $\bar{y}$.

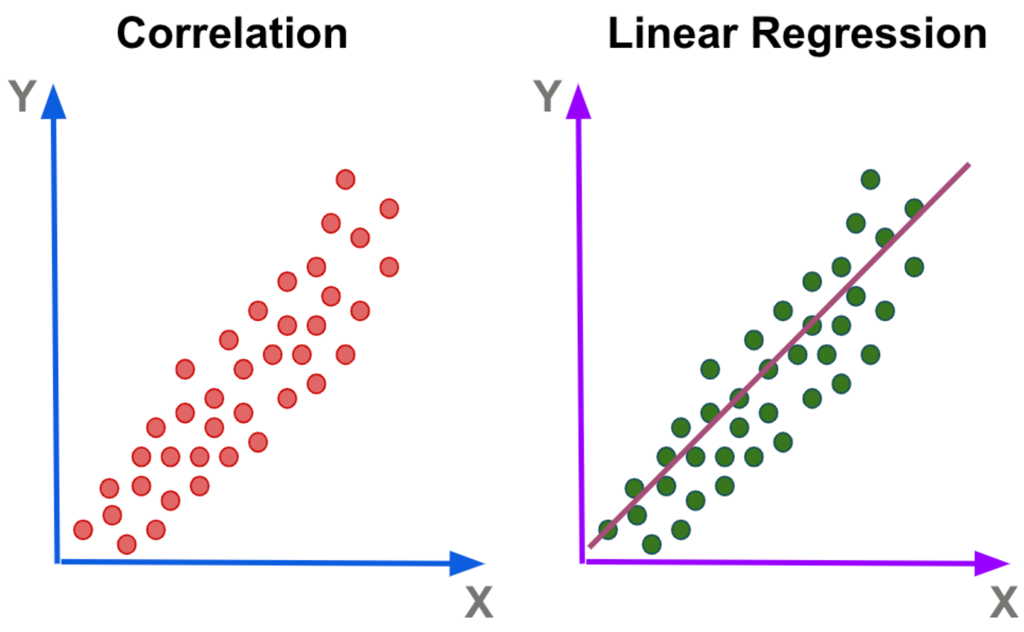
Correlation Coefficient

-1	-0.8	-0.6	0	0.6	0.8	1

A perfect negative relationship.	A strong negative relationship.	A moderate negative relationship.	No relationship.	A moderate positive relationship.	A strong positive relationship.	A perfect positive relationship.

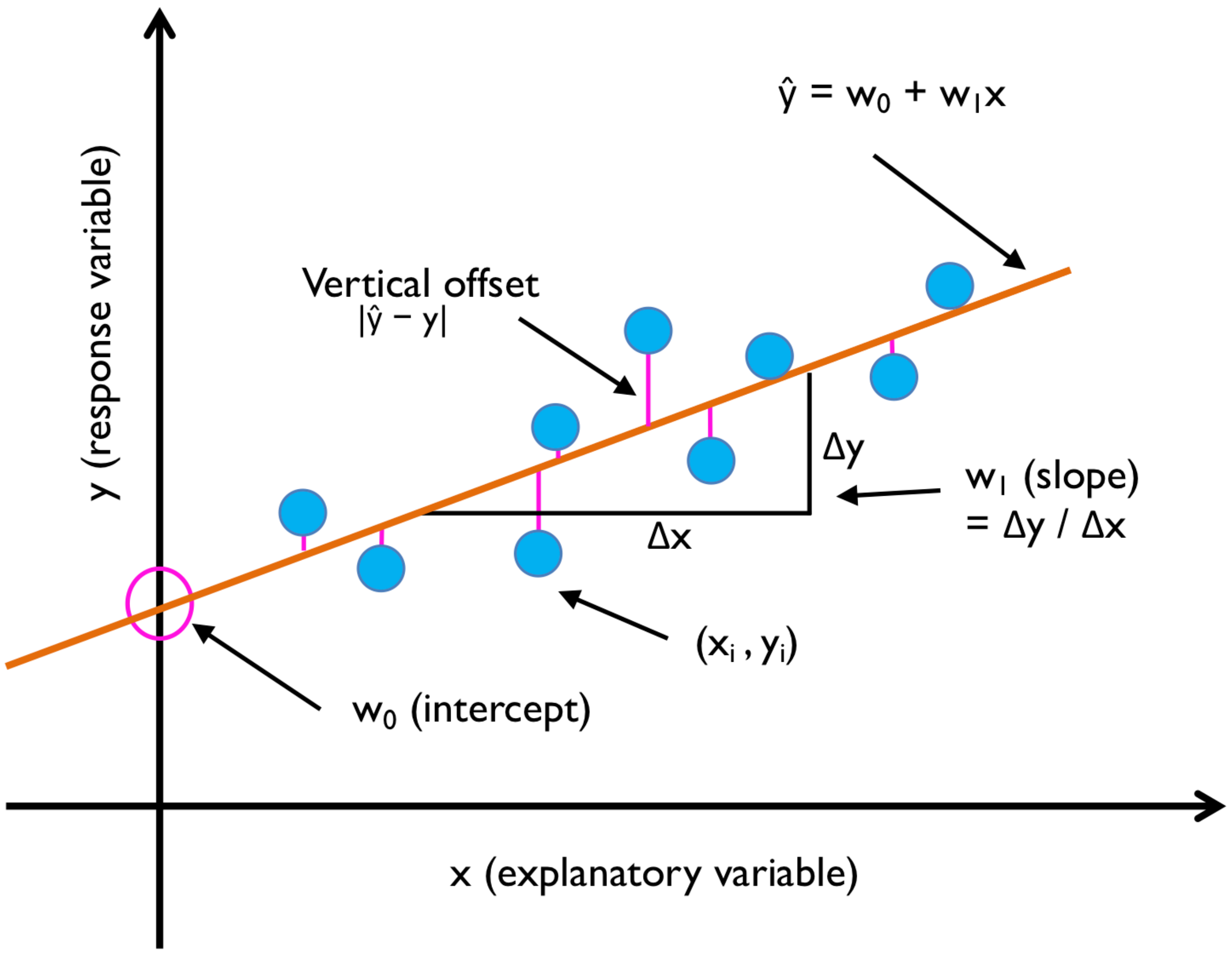
Correlation and Linear Regression Analysis

- Linear regression takes correlation analysis further and shows how much one variable affects another and, more importantly, whether you can use the pattern of one variable to predict and estimate the behaviour of another.



Correlation and Regression does not imply causation!!

Linear Regression



Linear Regression: Do we need arguments?

- In scikit-learn the LinearRegression library does not have arguments for finding the parameters of the model

```python
slr = LinearRegression()
slr.fit(X, y) # type: ignore
y_pred = slr.predict(X)
print('Slope: %.3f' % slr.coef_[0])
print('Intercept: %.3f' % slr.intercept_)
```

[28]

```
...
Slope: 9.102
Intercept: -34.671
```

Mathematical Proofs of Ordinary Least Squares

Objective: Find θ, such that : $y = X\theta$

Shapes: $y \rightarrow (n, 1)$ $X \rightarrow (n, m)$ $\theta \rightarrow (m, 1)$

Proof #1

1) Linear Regression Equation:

$$y = X\theta$$

2) We cannot invert X to get θ

$$\theta = X^{-1}y$$

...this is because X may not be square. Hence, non-invertible.

3) Multiple both sides of (1) by X^T

$$X^Ty = X^TX\theta$$

X^TX is square. Hence, invertible*.

4) Invert X^TX

$$\boxed{\theta = (X^TX)^{-1}X^Ty}$$

*In case of perfect multicollinearity, this won't be invertible.

Proof #2

1) Minimize Squared Error:

$$L = ||y - X\theta||^2$$

2) Split the squared norm into 2 terms

$$L = (y - X\theta)^T(y - X\theta)$$
$$= y^Ty - y^TX\theta - (X\theta)^Ty + (X\theta)^TX\theta$$

3) Minimize by differentiating w.r.t θ and then solve for θ

$$\frac{dL}{d\theta} = -2X^Ty + 2X^TX\theta$$

4) Set the derivative to zero

$$-2X^Ty + 2X^TX\theta = 0$$

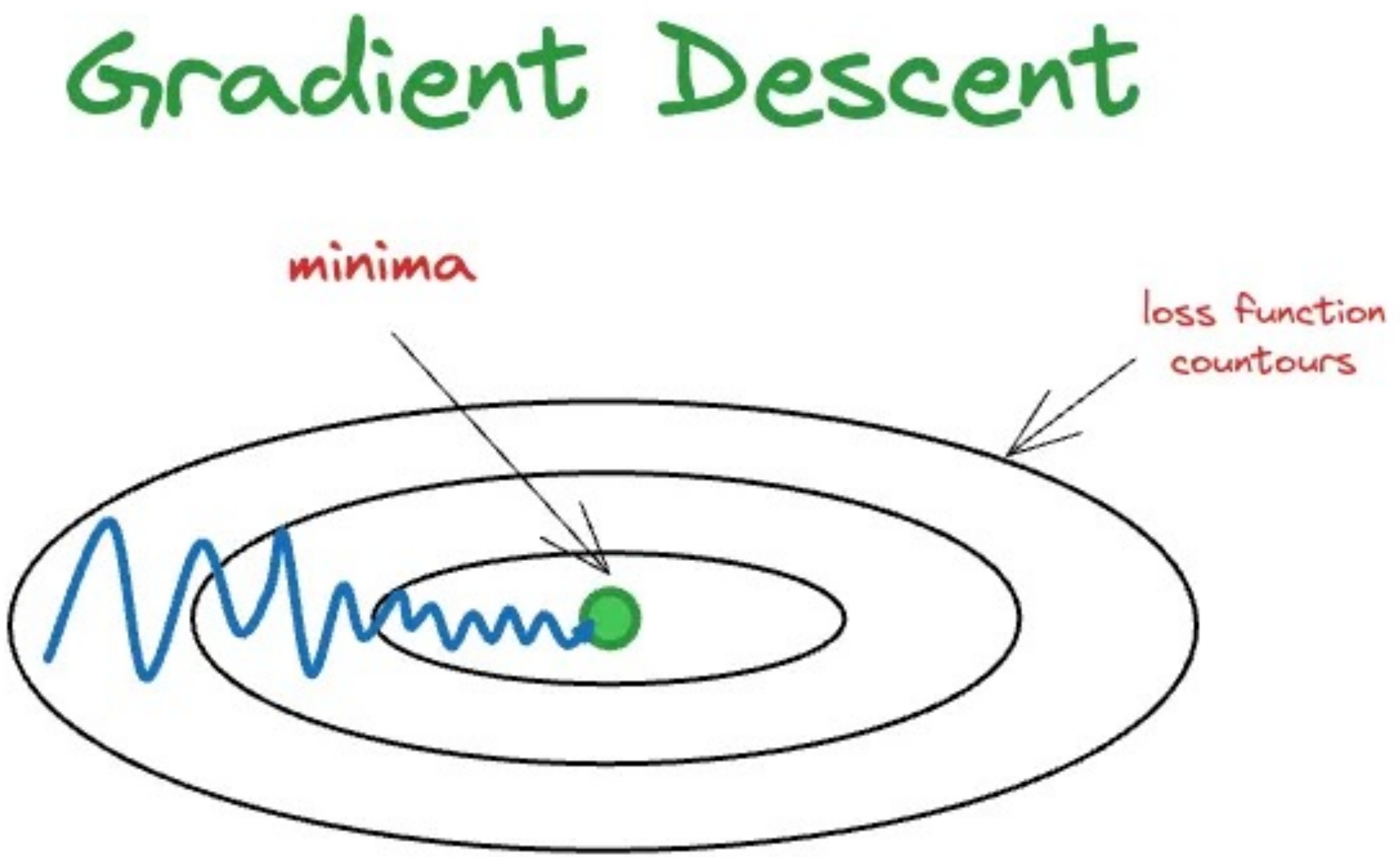
rearrange...

$$X^TX\theta = X^Ty$$

5) Invert X^TX

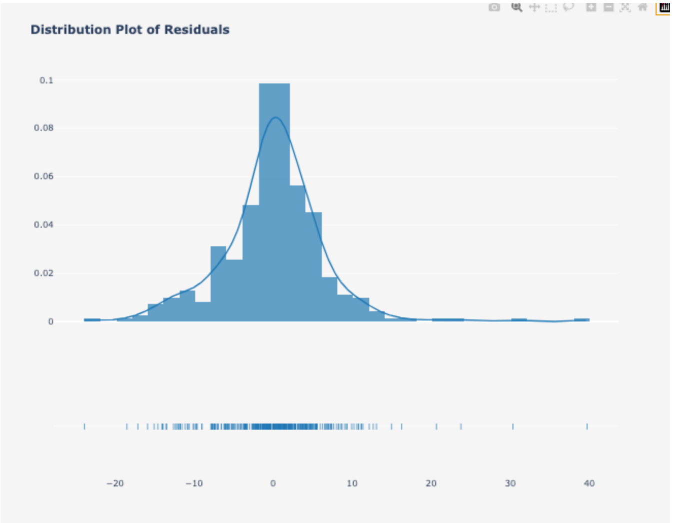
$$\boxed{\theta = (X^TX)^{-1}X^Ty}$$

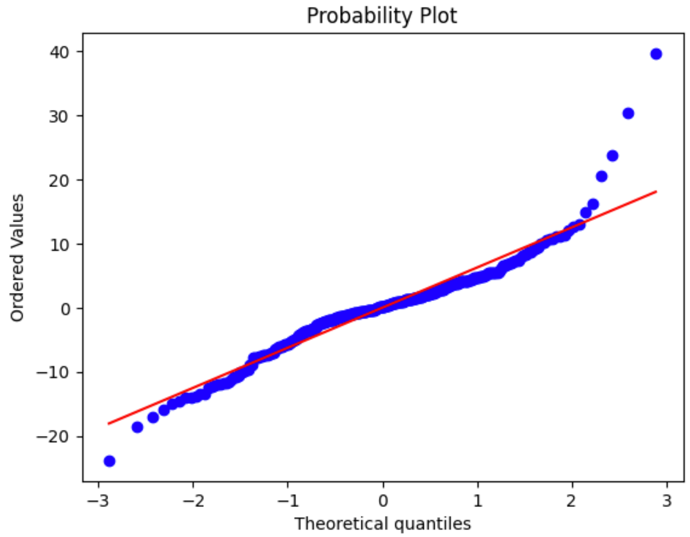
Linear Regression solved iteratively: Gradient Descent (GD)



Evaluating Linear Regression models

- Residual plots. Q-Q plots.





- MSE, RMSE and F-statistic for predictive power. Coefficient of determination (R^2) for variability explained.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}\left(y^{(i)} - \hat{y}^{(i)}\right)^2$$

$$R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{MSE}{Var(y)}$$

Playing with Linear Regression models: the Housing dataset and the Diabetes dataset

There are 2 problems in the Housing data:

- Racism: There is a great article, which was also cited in the Scikit-Learn documentation by M. Carlisle. It focuses on the main issues of the Boston Housing dataset, which he found that house prices effected by neighbourhood race.

- No suitable goal: "the goal of the research that led to the creation of this dataset was to study the impact of air quality but it did not give adequate demonstration of the validity of this assumption."